CRITICAL MINERALS · STRATEGIC ANALYSIS

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The Diversification Trap

65% of lithium processing runs through China. International agreements don't change that.

Market Access X

The Diversification Trap

Why Critical Minerals Agreements Won't Solve Anybody's Supply Chain Problems

Washington has signed more critical minerals agreements in the past three years than in the preceding three decades. The February 2026 Critical Minerals Ministerial — 54 countries, 21 bilateral frameworks, the launch of FORGE as the MSP's successor, and a $12 billion US stockpiling initiative — is the most visible escalation yet. The agreements are real. So is the gap between what they announce and what they can actually deliver. That gap is where supply chain risk actually concentrates.

This report is not investment advice. It is intended for professionals operating at the intersection of trade policy, geopolitics, and supply chain strategy.

Executive Summary

The dominant policy narrative conflates two things that need to be kept rigorously separate: upstream mining, which is genuinely diversifying, and midstream processing, which is not. The claim that US-led critical minerals diplomacy is fragmenting China's supply chain dominance does not survive contact with the second of these realities. China's control of chemical conversion, refining, and battery-grade material production is structurally entrenched and largely unaffected by the current generation of agreements, however diplomatically ambitious

— **The upstream is diversifying.** Mining geography is genuinely broadening across Australia, Latin America, and Africa. But this is a necessary, not sufficient, condition for supply chain independence.

— **The midstream is not.** China controls 65% of lithium chemical processing, 85%+ of rare earth separation, and approximately 80% of graphite anode material production. No allied project currently under development materially changes these figures before 2027 at the earliest, which is admittedly on the more optimistic side of most assessments.

— **The implementation gap is structural, not incidental.** FORGE, Project Vault, and the February Ministerial frameworks are the most serious allied response to date. But they do nothing to shorten the decade-plus timeline required to build processing ecosystems that rival China's.

— **China's primary risk is miscalibration, not displacement.** The more durable threat to China's position is not new mining capacity elsewhere — it is the gradual institutionalization of demand-side controls (procurement rules, subsidy eligibility, traceability standards) that make non-Chinese sourcing architectures path-dependent and difficult to reverse.

— **The compliance gap is real and underpriced.** Procurement rules demanding non-Chinese sourcing are arriving faster than non-Chinese processing capacity. That gap — between regulatory requirement and operational supply — is where your exposure concentrates.

Key Findings

1. Diplomatic velocity is real. Processing substitution is not.

US critical minerals diplomacy has accelerated dramatically across three successive administrations. The alliance architecture was institutionalized during the Biden administration, with the Minerals Security Partnership, the US-Japan IPEF minerals track, bilateral Critical Minerals Agreements with Australia, the UK, and a succession of resource-holding nations. The second Trump administration has sharpened the urgency and hardened the geopolitical framing of these efforts, while stripping away environmental constraints. The result has been a denser pipeline of agreements in a compressed timeframe.

The February 2026 Critical Minerals Ministerial represents the apex of this effort to date. FORGE — the Forum on Resource Geostrategic Engagement, announced as the MSP's successor — is a more ambitious construct than its predecessor: it proposes coordinated price floors maintained through adjustable tariffs, commits signatories to identifying deliverable projects within six months, and explicitly targets processing and refining stages rather than upstream mining alone. Vice President Vance's framing of 'reference prices for critical minerals at each stage of production' is the clearest signal yet that Washington has correctly identified where the chokepoint actually sits, namely in the midstream, not the mine.

The analytical question is not whether this ambition is genuine. It is whether frameworks, even when they are backed by $12 billion in stockpiling capital and a coordinated tariff architecture, are going to be enough to close a processing gap that took China approximately three decades to build.

65%	85%+	~80%	2027–30
Lithium chemicals processed in China	Global REE separation capacity in China	Graphite anode material, China-sourced	Earliest meaningful non-Chinese processing output

2. The implementation gap is structural, not incidental.

Mining and processing projects at the frontier of the diversification push require capital outlays measured in billions, timelines measured in decades, and technical ecosystems that cannot be assembled by policy memo. Rare earth separation facilities, battery-grade lithium conversion plants, and graphite anode processing installations are not analogous to semiconductor fabs or EV battery gigafactories, where speed of scale-up has been demonstrated. They require the accumulation of process knowledge that is largely tacit, embedded in experienced workforces and refined through years of production learning.

The earliest plausible commissioning dates for allied-led projects in REE separation and lithium chemical conversion extend to 2027 at the optimistic end, which is based on the questionable assumption that these projects suffer no permitting delays, no financing gaps, and remain insulated from market-cycle disruption to their investment cases. For rare earth separation at meaningful commercial scale outside China, the realistic timeline extends well into the 2030s.

In the United States, the judiciary remains a structural constraint independent of executive priorities. Nearly every high-profile critical minerals project the Trump administration has championed, from Resolution Copper, to McDermitt, to Silver Peak carries active or anticipated legal exposure from environmental review and tribal consultation litigation. The gap between project announcement and production reality is not primarily a political problem. It is an engineering, capital, and institutional problem, and no executive order resolves it within a commercially relevant timeframe.

> *The gap between diplomatic activity and material capacity is not a failure of political will. It is a function of physics, chemistry, and capital markets — and it runs to at least a decade.*

3. China's processing dominance is structurally entrenched.

China's advantage in critical minerals processing is not primarily a function of resource access. It is a function of accumulated know-how, industrial system integration, and the ability to deploy unmatched scale within a single national ecosystem. In rare earth separation, the relevant advantage is largely tacit: decades of process optimization embedded in workforce skills, facility design, and production routines that are not accessible through capital investment alone.

In battery-grade lithium chemicals and graphite anode materials, China's edge lies in yield optimization across tightly integrated industrial ecosystems. These advantages compound over time as production volumes accumulate. The structural logic is analogous to Apple's manufacturing entanglement with China: firms concentrated supply chains in a single geography because that geography was the only one capable of assembling the required industrial depth at scale. The same dynamic applies to processing-intensive mineral value chains.

This does not mean China's position is permanent. It means the timeline for material substitution is measured in decades rather than policy cycles. What's more, the current generation of agreements does not yet touch the processing stages where China's leverage actually concentrates.

4. The most underestimated risk: a strategic price war.

> ⚠ **CALIBRATION RISK**
>
> Chinese state-linked producers expanding output at sustained below-cost prices, timed to coincide with the commissioning of new Western processing facilities, would exploit the most vulnerable moment in any competitor's development cycle. Most Western diversification strategies implicitly assume rational market pricing will persist. That assumption has not been stress-tested against a coordinated Chinese response — and FORGE's price floor mechanism, while theoretically addressing this, has not yet been operationalized.

China retains the institutional capacity to tolerate prolonged losses in strategically important segments, constituting a core asymmetric advantage that Western private-sector competitors cannot match. A price war timed to undermine allied capacity build-out would be most damaging precisely when new projects are most financially vulnerable: the pre-ramp phase, when capital is committed but revenue is not yet flowing. The financing terms for most current diversification projects do not account for this scenario.

5. Demand-side governance shifts may prove more durable than supply-side agreements.

The demand-side governance shift is already in motion, and it is more analytically significant than the supply-side agreements that attract the most attention. When critical minerals are reframed as strategic inputs rather than ordinary commodities, and embedded in procurement rules, subsidy eligibility criteria, traceability requirements, and security standards, then buyer behavior becomes path-dependent. The qualification of alternative materials, the redesign of product specifications, and the reconfiguration of supplier relationships all constitute adjustments that do not get unwound simply because Chinese materials remain cheaper.

This creates a paradox for China. Adaptive responses that restore volume may fail to restore bargaining power. A supplier that is physically dominant but structurally excluded from procurement frameworks governing defense, EV, and energy system manufacturing in the US, EU, Japan, and South Korea has less leverage than the raw market share figure implies. The relevant question is not whether Chinese materials remain available, since they inevitably will. Instead, it is whether access to Western markets increasingly requires supply chain architectures that structurally exclude Chinese-controlled processing.

Forward-Looking Implications

The following horizon assessment distinguishes between developments that are already locked in, those contingent on execution against announced plans, and those dependent on political and market variables that remain genuinely uncertain.

Near Term (0–18 months)	Compliance pressure intensifies before supply alternatives exist. IRA and CRMA eligibility rules will continue tightening, with procurement standards in defense and EV sectors increasingly specifying non-Chinese-controlled sourcing. Processing capacity outside China will not meaningfully expand in this window. The compliance gap widens. FORGE's price floor mechanism — if operationalized — could begin stabilizing the investment case for allied processing projects, but operational details remain undefined. Expect China to accelerate export control pressure on gallium, germanium, and rare earth processing chemicals as a counter-leverage response.
Medium Term (18 months–5 years)	The first wave of allied processing capacity reaches commissioning — primarily Australian lithium hydroxide and Chilean lithium chemicals qualifying under existing FTAs. These are meaningful for lithium but leave rare earth separation and graphite anode processing almost entirely China-dependent through this period. FORGE's success or failure as a functioning plurilateral architecture will become evident: either it achieves coordinated pricing discipline and project delivery, or it fragments along the same lines as the MSP. China is most likely to deploy a strategic price response in this window, targeting projects at their most financially exposed pre-revenue phase.
Long Term (5+ years)	Material substitution in processing becomes structurally possible but not assured. The question shifts from 'can alternative capacity exist?' to 'is it commercially sustainable at scale without permanent policy protection?' Demand-side governance shifts — procurement rules, traceability requirements, buyer qualification standards — become the dominant force shaping market structure, regardless of whether Chinese supply remains cheaper. If buyer qualification processes have locked in non-Chinese supply chains during the medium-term window, those architectures persist. If they have not, the cost differential reasserts itself and Chinese processing dominance continues by default.

Actionable Takeaways

Each of the following is a concrete action, not a general orientation. The underlying logic applies regardless of where in an organization these questions are being asked, or what the strategic goals of a given market participant may be.

01 Stop treating project announcements as supply chain risk mitigation.

An announced project is not a supply alternative — it is a pipeline entry. The relevant milestone for planning purposes is commissioning date for processing and chemical conversion capacity specifically, not mining. Most current announcements refer to the latter. Very few refer to the former.

02 Map your exposure by processing stage, not by mineral.

The relevant question is not which mineral a firm is exposed to but at which stage of the value chain its supply is sourced — and whether that stage passes through Chinese-controlled processing. A supply chain that sources mined lithium from Australia but converts it to battery-grade hydroxide in China is not diversified in any operationally meaningful sense. Audit tier-2 and tier-3 suppliers accordingly.

03 Get ahead of compliance definitions before they are finalized.

The IRA's foreign entity of concern provisions and the EU's CRMA sourcing thresholds are still being translated into operational procurement criteria. The difference between 'Chinese-processed' and 'Chinese-controlled' is not semantic — it determines whether large swaths of current supply chains remain compliant. The window to influence those definitions is closing.

04 Build FORGE and price floor operationalization into your scenario planning.

If FORGE's price floor mechanism is operationalized — even for a subset of processing stages and member countries — it changes the investment calculus for allied processing capacity in ways that could meaningfully accelerate supply alternatives. Most supply chain scenario models currently ignore this. They should not.

05 Treat buyer qualification of non-Chinese material as a leading indicator, not a lagging one.

Qualification processes — the technical validation of alternative materials against product specifications — are the rate-limiting step between a processing project existing and one that matters. Monitor whether major downstream OEMs and defense primes are actively qualifying non-Chinese-sourced alternatives. Those that move early build optionality; those that do not risk being locked into non-compliant supply chains after qualification standards have set

06 Do not assume pricing rationality in the medium-term window.

The period between now and the first wave of allied processing capacity reaching meaningful scale is the window during which a Chinese strategic price response is most likely and could be most damaging. Diversification investments — whether direct, through offtake agreements, or through investment in processing projects — should be stress-tested against an extended period of below-cost Chinese pricing. Projects that are viable at current prices but not at 30–40% lower prices are not adequately de-risked.

Indicators to Watch

The following are the specific signals that will indicate whether the trajectory is shifting — and in which direction. These are operational intelligence triggers, not background monitoring items.

01 FORGE price floor architecture — Interagency rulemaking

The degree of specificity emerging from the interagency process — which minerals, which processing stages, which partner countries — will determine whether FORGE's price floor mechanism is operational policy or diplomatic language. The Federal Register is where formal action will eventually appear; the US-Mexico and US-EU-Japan Action Plans are the near-term indicators of how price floors will actually be structured and enforced.

02 IRA/CRMA 'foreign entity of concern' definition finalization

The gap between 'Chinese-processed' and 'Chinese-controlled' in subsidy eligibility rules is where significant compliance exposure will concentrate. Track regulatory drafting in Treasury (IRA) and the European Commission (CRMA) for how this distinction is being operationalized.

03 Processing commissioning dates — not announcement dates

Track commissioning schedules for battery-grade lithium hydroxide (Australia, Chile), REE separation (US, Australia, Estonia), and graphite anode material (non-China) specifically. Any material slip in these timelines extends the compliance gap.

04 Chinese SOE capital discipline signals

Enforced ROE targets, rising project cancellations framed as financial discipline, tightening policy bank credit to upstream and midstream materials — any of these signals that China's capacity to sustain counter-cyclical investment is under stress. That narrows the strategic price war risk.

05 OEM and defense prime qualification of non-Chinese material

Active qualification processes by Tesla, CATL's non-Chinese competitors, major defense primes, and energy system integrators are the leading indicator of whether demand-side governance shifts are translating into operational supply chain change.

06 Judicial deference to executive national-security permitting

Watch for Supreme Court language narrowing judicial review of executive national-security designations, and lower-court deference to security-critical project classifications. If permitting ceases to be a binding constraint in targeted US segments, project timelines compress — and the supply-side outlook improves faster than current models project.

The core conclusion of this report is analytically uncomfortable for both sides of the policy debate. Western diversification of critical minerals supply chains is more real, more institutionally serious, and more durable than skeptics acknowledge. It is also further from generating the processing-stage substitution that would actually alter China's structural position than advocates admit.

The risk that needs to be managed is not that diversification fails entirely. It is that it succeeds slowly enough to create a compliance gap between when procurement rules demand non-Chinese sourcing and when non-Chinese processing capacity actually exists at the scale and quality that a resilient and commercially viable supply chain requires. That gap is not yet being priced into most supply chain risk assessments. Pricing it in is the analytical work this report is intended to support.

SOURCES & REFERENCES

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